

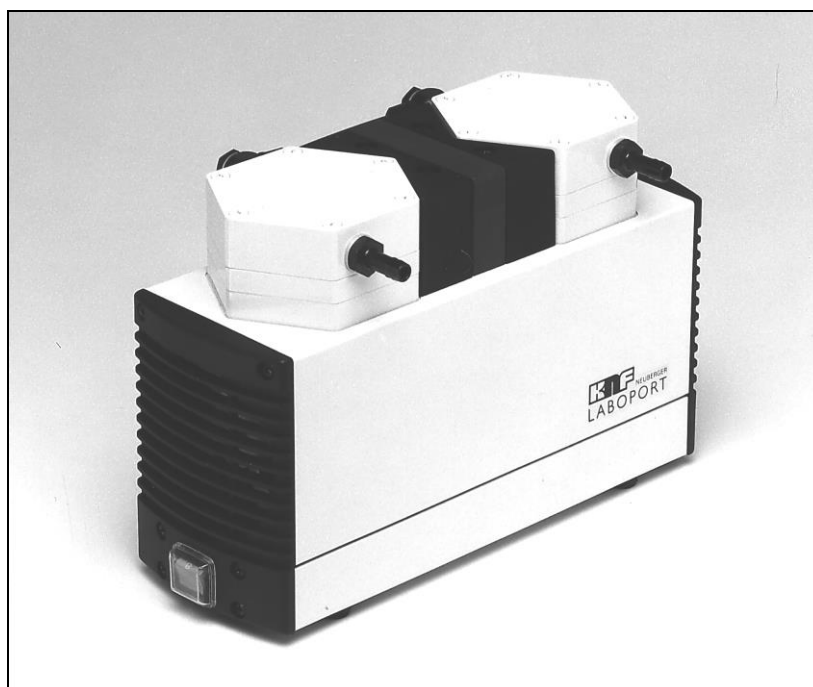
Operating Instructions

Read and observe these Operating Instructions!

Laboport® Laboratory Pumps

N 820 AN.18
N 820 AT.18

N 820.3 AN.18
N 820.3 AT.18



KNF Neuberger GmbH
Alter Weg 3
D-79112 Freiburg
Germany
Phone +49-(0)7664 / 5909-0
Fax +49-(0)7664 / 5909-99
E-Mail: info@knf.de
www.knf.de

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1. About this document

1.1. Using the Operating Instructions

The Operating Instructions are part of the pump.

- ➔ Carefully study the Operating Instructions before using a pump.
- ➔ Always keep the Operating Instructions handy in the work area.
- ➔ Pass on the Operating Instructions to the next owner.
- ➔ If you use a pump from the N 820.A_.18 or N 820.3 A_.18 range in a vacuum system, take care to observe the Operating Instructions for Vacuum Systems as well as these instructions.

Project pumps

Customer-specific project pumps (pump models which begin with “PJ” or “PM”) may differ from the Operating Instructions.

- ➔ For project pumps, also observe the agreed upon specifications.

1.2. Symbols and Markings

Warning



WARNING

A danger warning is located here.

Possible consequences of a failure to observe the warning are specified here. The signal word, e.g. Warning, indicates the danger level.

- ➔ Measures for avoiding the danger and its consequences are specified here.

Danger levels

Signal word	Meaning	Consequences if not observed
DANGER	warns of immediate danger	Death or serious injuries and/or serious damage are the consequence.
WARNING	warns of possible danger	Death or serious injuries and/or serious damage are possible.
CAUTION	warns of a possibly dangerous situation	Minor injuries or damage are possible.

Tab. 1

Other information and symbols

- ➔ An activity to be carried out (a step) is specified here.
- 1. The first step of an activity to be carried out is specified here. Additional, consecutively numbered steps follow.



This symbol refers to important information.

2. Use

2.1. Proper use

The pumps are exclusively intended for transferring gases and vapors.

Owner's responsibility

Operating parameters and conditions

Only install and operate the pumps under the operating parameters and conditions described in chapter 4, Technical data.

Make sure that the installation location is dry and the pump is protected against rain, splash, hose and drip water.

Requirements for transferred medium

Before using a medium, check whether the medium can be transferred danger-free in the specific application case.

Before using a medium, check the compatibility of the materials of the pump head, diaphragm and valves with the medium.

Only transfer gases which remain stable under the pressures and temperatures occurring in the pump.

Accessories

Laboratory equipment or additional components connected to a pump have to be suitable for use with the pneumatic capabilities of the pump.

2.2. Improper use

The pumps may not be operated in an explosive atmosphere.

The pumps are not suitable for transferring dusts.

The pumps are not suitable for transferring liquids.

The pumps must not be used to create vacuum and overpressure simultaneously.

An overpressure must not be applied to the suction side of the pump.

3. Safety

i Note the safety precautions in chapters 6. *Installation and connection*, and 7. *Operation*.

The pumps are built according to the generally recognized rules of technology and in accordance with the occupational safety and accident prevention regulations. Nevertheless, dangers can result during their use which lead to injuries to the user or others, or to damage to the pump or other property.

Only use the pumps when they are in a good technical and proper working order, in accordance with their intended use, observing the safety advice within the operating instructions, at all times.

Personnel

Make sure that only trained and instructed personnel or specially trained personnel work on the pumps. This especially applies to assembly, connection and servicing work.

Make sure that the personnel has read and understood the operating instructions, and in particular the "Safety" chapter.

Working in a safety-conscious manner

Observe the accident prevention and safety regulations when performing any work on the pump and during operation.

Do not expose any part of your body to the vacuum.

Open housing parts with notice sticker (see fig. 1) only after separating mains plug from power source.

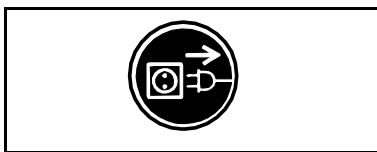


Fig. 1: Notice sticker

Handling dangerous media

When transferring dangerous media, observe the safety regulations when handling these media.

Handling flammable media

Be aware that the pumps are not designed to be explosion-proof.

Make sure the temperature of the medium is always sufficiently below the ignition temperature of the medium, to avoid ignition or explosion. This also applies for unusual operational situations.

Note that the temperature of the medium increases when the pump compresses the medium.

Hence, make sure the temperature of the medium is sufficiently below the ignition temperature of the medium, even when it is compressed to the maximum permissible operating pressure of the pump. The maximum permissible operating pressure of the pump is stated in the technical specifications (see chapter 4).

If necessary, consider any external sources of energy, such as radiation, that may add heat to the medium.

In case of doubt, consult the KNF customer service.

Environmental protection

Store all replacement parts in a protected manner and dispose of them properly in accordance with the applicable environmental protection regulations. Observe the respective national and international regulations. This especially applies to parts contaminated with toxic substances.

Standards	The pumps conform to the Directive 2011/65/EU (RoHS2). The pumps conform to the safety regulations of the EC Directive 2014/30/EU concerning Electromagnetic Compatibility and the EC Directive 2006/42/EC concerning Machinery. The following harmonized standards have been used: ▪ DIN EN 61010-1 ▪ DIN EN 61326-1 - class A ▪ DIN EN 50581 The pumps correspond to IEC 664: ▪ the overvoltage category II ▪ the pollution degree 2.
Customer service and repairs	Only have repairs to the pump carried out by the KNF Customer Service responsible. Only authorized personnel should open those parts of the housing that contain live electrical parts. Use only genuine parts from KNF for servicing work.

4. Technical Data

i All pumps are secured against overheating with thermal switches and are equipped with a mains fuse.

Pump materials N 820 AN.18 and N 820.3 AN.18

<i>Component</i>	<i>Pump material</i>
Pump head	Aluminum alloy
Diaphragm	CR
Valve	NBR

Tab. 2

Pump materials N 820 AT.18 and N 820.3 AT.18

<i>Component</i>	<i>Pump material</i>
Pump head	Aluminum alloy
Diaphragm	PTFE coated
Valve	FFPM

Tab. 3

i Refer to the type plate for the pump's electrical configuration.

N 820 AN.18

Pneumatic performance			
Max. permissible operating pressure [bar g]	1.0		
Ultimate vacuum [mbar abs.]	≤ 100		
Delivery rate at atm. pressure [l/min]*	max. 22		
Pneumatic connections			
Hose connection [mm]	ID 9		
Other parameters			
Weight [kg]	7.1		
Dimensions : L x H x W [mm]	261 x 204 x 110		
Maximum permissible ambient relative humidity	80 % for temperatures up to 31 °C, decreasing linearly to 50 % at 40 °C		
Max. altitude of site [m above sea level]	2000		
Ambient and media temperature			
Permissible ambient temperature	+ 5 °C to + 40 °C		
Permissible media temperature	+ 5 °C to + 40 °C		
Electrical Data			
Voltage [V]	100	115	230
Frequency [Hz]	50/60	60	50
Max. operating current [A]	2.2	1.9	0.9
Power consumption pump [W]	145	145	130
Maximum permitted mains voltage fluctuations	+/- 10 %	+/- 10 %	+/- 10 %
Fuse pump (2x) T [A]	4.0	3.15	2.0
Protection class motor	IP44		

Tab. 4

* Liters in standard state (1,013 mbar)

N 820 AT.18

<i>Pneumatic performance</i>			
Max. permissible operating pressure [bar g]	1.0		
Ultimate vacuum [mbar abs.]	≤ 100		
Delivery rate at atm. pressure [l/min]*	max. 20		
<i>Pneumatic connections</i>			
Hose connection [mm]	ID 9		
<i>Other parameters</i>			
Weight [kg]	7.1		
Dimensions : L x H x W [mm]	261 x 204 x 110		
Maximum permissible ambient relative humidity	80 % for temperatures up to 31 °C, decreasing linearly to 50 % at 40 °C		
Max. altitude of site [m above sea level]	2000		
<i>Ambient and media temperature</i>			
Permissible ambient temperature	+ 5 °C to + 40 °C		
Permissible media temperature	+ 5 °C to + 40 °C		
<i>Electrical Data</i>			
Voltage [V]	100	115	230
Frequency [Hz]	50/60	60	50
Max. operating current [A]	2.2	1.9	0.9
Power consumption pump [W]	145	145	130
Maximum permitted mains voltage fluctuations	+/- 10 %	+/- 10 %	+/- 10 %
Fuse pump (2x) T [A]	4.0	3.15	2.0
Protection class motor	IP44		

Tab. 5

* Liters in standard state (1,013 mbar)

N 820.3 AN.18

<i>Pneumatic performance</i>			
Max. permissible operating pressure [bar g]	1.0		
Ultimate vacuum [mbar abs.]	≤ 8		
Delivery rate at atm. pressure [l/min]*	max. 22		
<i>Pneumatic connections</i>			
Hose connection [mm]	ID 9		
<i>Other parameters</i>			
Weight [kg]	9.3		
Dimensions : L x H x W [mm]	304 x 204 x 110		
Maximum permissible ambient relative humidity	80 % for temperatures up to 31 °C, decreasing linearly to 50 % at 40 °C		
Max. altitude of site [m above sea level]	2000		
<i>Ambient and media temperature</i>			
Permissible ambient temperature	+ 5 °C to + 40 °C		
Permissible media temperature	+ 5 °C to + 40 °C		
<i>Electrical Data</i>			
Voltage [V]	100	115	230
Frequency [Hz]	50/60	60	50
Max. operating current [A]	1.8	1.2	0.7
Power consumption pump [W]	130	130	120
Maximum permitted mains voltage fluctuations	+/- 10 %	+/- 10 %	+/- 10 %
Fuse pump (2x) T [A]	3.15	2.5	1.6
Protection class motor	IP44		

Tab. 6

* Liters in standard state (1,013 mbar)

N 820.3 AT.18

Pneumatic performance			
Max. permissible operating pressure [bar g]	1.0		
Ultimate vacuum [mbar abs.]	≤ 8		
Delivery rate at atm. pressure [l/min]*	max. 20		
Pneumatic connections			
Hose connection [mm]	ID 9		
Other parameters			
Weight [kg]	9.3		
Dimensions : L x H x W [mm]	304 x 204 x 110		
Maximum permissible ambient relative humidity	80 % for temperatures up to 31 °C, decreasing linearly to 50 % at 40 °C		
Max. altitude of site [m above sea level]	2000		
Ambient and media temperature			
Permissible ambient temperature	+ 5 °C to + 40 °C		
Permissible media temperature	+ 5 °C to + 40 °C		
Electrical Data			
Voltage [V]	100	115	230
Frequency [Hz]	50/60	60	50
Max. operating current [A]	1.8	1.2	0.7
Power consumption pump [W]	130	130	120
Maximum permitted mains voltage fluctuations	+/- 10 %	+/- 10 %	+/- 10 %
Fuse pump (2x) T [A]	3.15	2.5	1.6
Protection class motor	IP44		

Tab. 7

* Liters in standard state (1,013 mbar)

5. Design and Function

Design

- 1 Connection piece
- 2 Pneumatic connection
- 3 Pump head
- 4 Outlet (pressure side)
- 5 Inlet (suction side)
- 6 Power switch

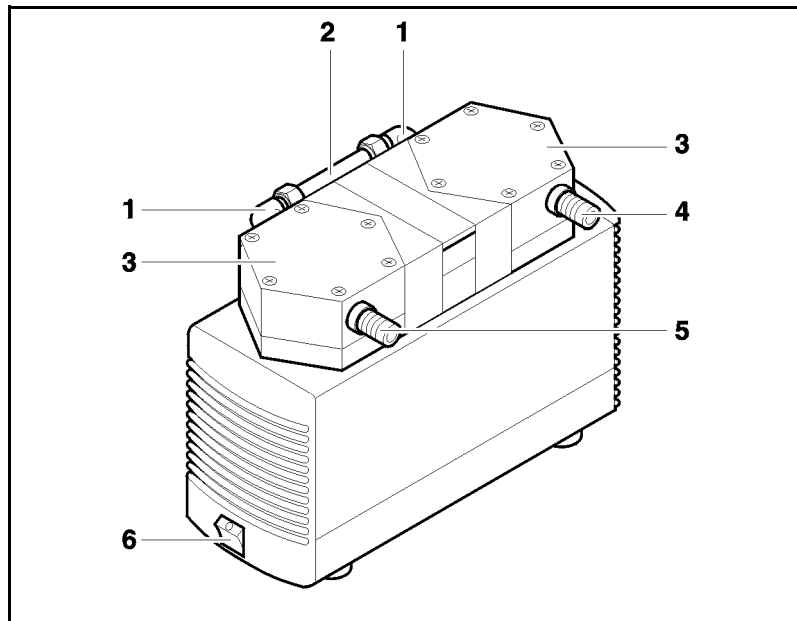


Fig. 2: Diaphragm pump (shown: pump N 820.3 AT.18, symbolic)

Function Diaphragm Pump

- 1 Outlet valve
- 2 Inlet valve
- 3 Transfer chamber
- 4 Diaphragm
- 5 Eccentric
- 6 Connecting rod
- 7 Pump drive

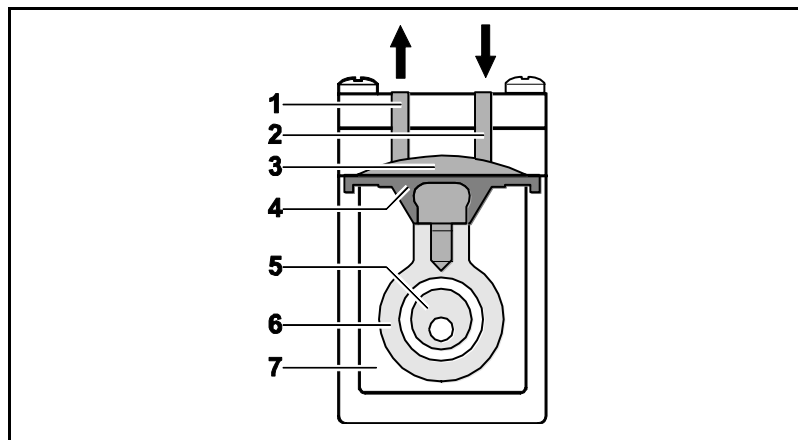


Fig. 3: Pump head

Diaphragm pumps transfer, compress (depending on pump version) and evacuate gases and vapors.

The elastic diaphragm (4) is moved up and down by the eccentric (5) and the connecting rod (6). In the downward stroke it aspirates the gas to be transferred via the inlet valve (2). In the upward stroke, the diaphragm presses the medium out of the pump head via the outlet valve (1). The transfer chamber (3) is hermetically separated from the pump drive (7) by the diaphragm.

6. Installation and connection

Only install the pumps under the operating parameters and conditions described in chapter 4, Technical data.

Observe the safety precautions (see chapter 3).

6.1. Installation

- ➔ Before installation, store the pump at the installation location to bring it up to room temperature.
- Dimensions ➔ See chapter 4, Technical data, for the dimensions of pump.
- Cooling air supply ➔ Install the pump so that the motor fan can intake sufficient cooling air.
- Installation location ➔ Make sure that the installation location is dry and the pump is protected against rain, splash, hose and drip water.
- ➔ Choose a safe location (flat surface) for the pump.
- ➔ Protect the pump from dust.
- ➔ Protect the pump from vibration and jolt.

6.2. Connection

- Connected components ➔ Only connect components to the pump which are designed for the pneumatic data of the pump (see chapter 4).
- Pump exhaust ➔ If the pump is used as a vacuum pump, safely discharge the pump exhaust at the pump's pneumatic outlet.
- Connection **i** A marking on the pump head shows the direction of flow.
 1. Remove the protective plugs from the pneumatic connectors of the pump.
 2. Connect the suction line and pressure line (tube: inner diameter 9 mm).
 3. Lay the suction and pressure line at a downward angle to prevent condensate from running into the pump.
 4. Insert the power cable's plug into a properly installed shockproof socket.

7. Operation

7.1. Preparing for Start-up

Before switching on the pump, observe the following points:

	Operational requirements
Pump	▪ All hoses attached properly ▪ Fan openings not blocked ▪ Specifications of the power supply correspond with the data on the pump's type plate. ▪ The pump outlet is not closed or constricted.

Tab. 8

7.2. Starting

- ➔ Only operate the pump under the operating parameters and conditions described in chapter 4, Technical data.
- ➔ Make sure the pump is used properly (see chapter 2.1).
- ➔ Make sure the pump is not used improperly (see chapter 2.2).
- ➔ Observe the safety precautions (see chapter 3).



WARNING

Hazard of the pump head bursting due to excessive pressure increase

- ➔ Do not exceed max. permissible operating pressure (see chapter 4).
- ➔ Monitor pressure during operation.
- ➔ If the pressure exceeds the maximum permissible operating pressure, immediately shut down pump and eliminate fault (see chapter 9. Troubleshooting).
- ➔ Only throttle or regulate the air or gas quantity in the suction line to prevent the maximum permissible operating pressure from being exceeded.
- ➔ If the air or gas quantity in the pressure line is throttled or regulated, make sure that the maximum permissible operating pressure of the pump is not exceeded.

i Excessive pressure (with all of the related hazards) can be prevented by placing a bypass line with a pressure-relief valve between the pressure and suction side of the pump. For further information, contact your KNF technical adviser.

- Pump standstill ➔ With the pump at a standstill, open pressure and suction lines to normal atmospheric pressure.

**WARNING**

Automatic starting can cause personal injury and pump damage

When the operation of the pump is interrupted by the thermal switch, the pump will restart automatically after cooling down.

- ➔ After triggering of the thermal protection or in the event of power failure, remove the pump's mains plug from the socket so that the pump cannot start uncontrollably.
 - ➔ Attempt work on the pump only if the pump is separated from mains power.
-

7.3. Switching pump on and off

Switching pump on

i The pump may not start up against pressure during switch-on. This also applies in operation following a brief power failure. If a pump starts against pressure, it may block. This activates the thermal switch, and the pump switches off.

- ➔ Make sure that no pressure is present in the lines during switch-on.
- ➔ Switch on pump with mains switch (see fig. 2).

Switching off the pump/removing from operation

- ➔ When transferring aggressive media, flush the pump prior to switch-off to increase the service life of the diaphragm (see chapter 8.2.1).
- ➔ Switch off pump with mains switch (see fig. 2).
- ➔ Open pressure and suction lines to normal atmospheric pressure.
- ➔ Disconnect the power source.

8. Servicing

8.1. Servicing Schedule

Component	Servicing interval
Pump	Regular inspection for external damage or leaks
Diaphragm and valve plates/ sealings	Replace at the latest, when pump output decreases

Tab. 9

8.2. Cleaning

i When cleaning, make sure that no liquids enter the inside of the housing.

8.2.1. Flushing Pump

➔ Before switching off the pump, flush it with air (if necessary for safety reasons: with an inert gas) for about five minutes under atmospheric conditions (ambient pressure).

8.2.2. Cleaning Pump

- ➔ Only use solvents for cleaning if the head materials cannot be attacked (check the resistance of the material!).
- ➔ If compressed air is available, blow out the components.

8.3. Changing Diaphragm and Valves

- Conditions
- Pump is switched off and mains plug is removed from the socket
 - Pump is clean and free of hazardous materials
 - Tubes removed from pump's pneumatic inlet and outlet
 - Pump is removed from baseplate (if pump is integrated into a system)

Spare parts/tools

Spare part/tool
Service Set (according to chapter 10)
Philips-head screwdriver No. 2
Felt-tip pen

Tab. 10

- Information on procedure
- ➔ Always replace diaphragm and valve plates/sealings together to maintain the pump performance.
- With multi-head pumps, parts of the individual pump heads can be confused.
- ➔ Replace the diaphragm and valve plates/sealings of the individual pump heads consecutively.



WARNING

Health hazard due to dangerous substances in the pump!

Depending on the substance transferred, caustic burns or poisoning are possible.

- ➔ Wear protective clothing if necessary, e.g. protective gloves.
- ➔ Flush pump before replacing the diaphragm and valve plates/sealings (see chapter 8.2.1).

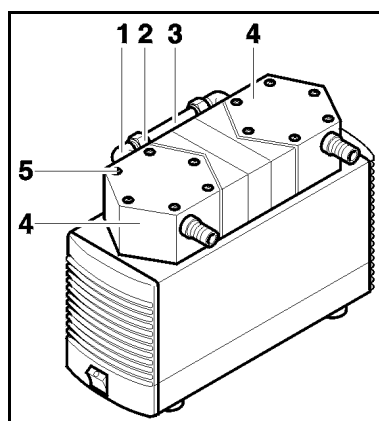


Fig. 4: Removing pump head

Removing pump head

1. For two-headed pumps: On the pneumatic head connection (3), loosen the union nuts (2) by hand. Then slightly loosen the angle-fitting (1) in the pump head (4) by turning it counter-clockwise, so that the connecting tube can be pulled out.
2. Mark the position of head plate (fig. 5/2), intermediate plate (fig. 5/4) and adapter (fig. 5/10) relatively to each other by a drawing line (for two-headed pumps: at both pump heads) with a felt-tip pen. This helps to avoid incorrect assembly later.
3. Loosen the outer screws (5) on the pump head/pump heads.
4. Carefully remove pump head / pump heads.

Change diaphragm

i Replace the diaphragms of two-headed pumps consecutively in order to ensure that the same number of diaphragm spacers is used.

1. For two-headed pumps: Push down one diaphragm until other diaphragm is pushed upwards to its highest position.
2. Carefully unscrew the upper diaphragm (fig. 5/6) counter-clockwise using both hands.
3. Shift spacer thick (7) and spacers thin (8) from the old diaphragm onto the screw thread of the new diaphragm (same number and order).
4. Screw in the new diaphragm and tighten it by hand.
5. For two-headed pumps: Carry out steps 1 to 4 for the second pump head.

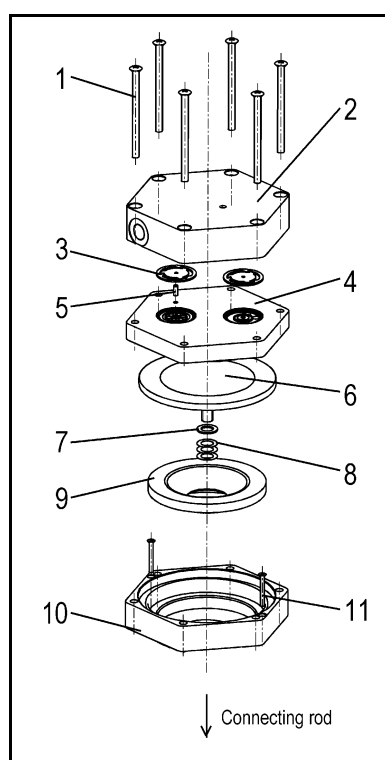


Fig. 5: Pump head

Specification

- | | |
|-----------|---------------------|
| 1 | Screw |
| 2 | Head plate |
| 3 | Valve plate/sealing |
| 4 | Intermediate plate |
| 5 | Pin guide |
| 6 | Diaphragm |
| 7 | Spacer (thick) |
| 8 | Spacer (thin) |
| 9 | Dampening ring |
| 10 | Adapter |
| 11 | Screw |

Change valve plates/sealings

i Replacing the valve plates/sealings of two-headed pumps consecutively.

1. Remove head plate (2) from intermediate plate (4). Valve plates/sealings (3) are visible.
2. Remove old valve plate/sealings.
3. Clean intermediate plate (4) carefully (if there should be deposits in the recesses in the intermediate plate).
4. Insert new valve plates/sealings (3) in the recesses in the intermediate plate (4).
5. For two-headed pumps: Carry out the steps 1 to 4 for the second pump head.
6. Dispose of the old diaphragm and valve plates/sealings properly.

Refitting pump head

1. Apply pressure all around the edge of the diaphragm. With a two-headed pump: Apply pressure to diaphragm on only one pump head.
2. Place the intermediate plate (4) with valve plates/sealings (3) on the adapter (10) in accordance with the felt-tip pen marking.
3. Place the head plate (2) on the intermediate plate (4) in the position indicated by the guide pin (5).
4. Gently tighten screws (1) in diagonal order.
5. For two-headed pumps: Carry out steps 1 to 4 for the second pump head.
6. For two-headed pumps: Refit the pneumatic head connection: Place tube onto the connecting part of the angle fitting, turn angle fitting to a straight position and tighten the union nut.

Final steps

1. Remount the pump to the baseplate (if applicable).
2. Reconnect suction and pressure line to the pump.
3. Reconnect the pump to the electricity supply.

9. Troubleshooting



Extreme danger from electrical shock!

→ Disconnect the pump power supply before working on the pump.

DANGER

→ Make sure the pump is de-energized and secure.

→ Check the pump (see Tab. 11 to 14).

Pump does not transfer	
Cause	Fault remedy
No voltage in the power source	→ Check room fuse and switch on if necessary.
Thermal switch has operated following to over-heating.	→ Disconnect pump from mains. → Allow pump to cool. → Trace cause of over-heating and eliminate it.
Connections or lines blocked.	→ Check connections and lines. → Remove blockage.
External valve is closed or filter is clogged.	→ Check external valves and filters.
Condensate has collected in pump head.	→ Detach the condensate source from the pump. → Flush pump (see chapter 8.2.1).
Diaphragm or valve plates/sealings are worn.	→ Replace diaphragm and valve plates/sealings (see chapter 8.3).

Tab. 11

Flow rate, pressure or vacuum too low	
The pump does not achieve the output specified in the Technical data or the data sheet.	
Cause	Fault remedy
Condensate has collected in pump head.	→ Detach the condensate source from the pump. → Flush pump (see chapter 8.2.1).
There is gauge pressure on pressure side and at the same time vacuum or a pressure above atmospheric pressure on suction side.	→ Change the pressure conditions.
Pneumatic lines or connection parts have an insufficient cross section.	→ Disconnect pump from system to determine output values. → Eliminate throttling (e.g. valve) if necessary. → Use lines or connection parts with larger cross section if necessary.
Leaks occur on connections, lines or pump head.	→ Check that tubes sit correctly on hose nozzles. → Replace leaky tubes. → Eliminate leaks.
Connections or lines completely or partially clogged.	→ Check connections and lines. → Remove the clogging parts and particles.
Head parts are soiled.	→ Clean head components.
Diaphragm or valve plates/sealings are worn.	→ Replace diaphragm and valve plates/sealings (see chapter 8.3).

Flow rate, pressure or vacuum too low	
The pump does not achieve the output specified in the Technical data or the data sheet.	
Cause	Fault remedy
Replaced diaphragms and valve plates/sealings	➔ Check that the spacers have been replaced onto the diaphragm screw thread. ➔ Check head connection and hose connections for leaks. ➔ Possibly carefully tighten the outer screws (fig. 5/1) of the top plate crosswise.

Tab. 12

Pump is switched on, but does not run, the on/off-switch on the pump is not lit	
Cause	Fault remedy
Pump is not connected with the power source.	➔ Connect pump to mains power.
No voltage in the power source	➔ Check room fuse and switch on if necessary.
Fuse in the pump is defective.	➔ Remove pump's mains plug from the socket. ➔ Loosen marked lid on underside of the pump. ➔ Select and replace suitable fuse (see chapter 4).

Tab. 13

Pump is switched on, but does not run, the on/off-switch on the pump is lit	
Cause	Cause
The thermal switch has opened due to overheating	➔ Remove pump's mains plug from the socket. ➔ Allow pump to cool. ➔ Trace cause of over-heating and eliminate it.

Tab. 14

Fault cannot be rectified

If you are unable to determine any of the specified causes, send the pump to KNF Customer Service (see last page for the address).

1. Flush the pump to free the pump head of dangerous or aggressive gases (see chapter 8.2.1).
2. Clean the pump (see chapter 8.2.2).
3. Send the pump, together with completed Health and Safety Clearance and Decontamination Form (Chapter 12), to KNF stating the nature of the transferred medium.

10. Spare parts and accessories

10.1. Spare parts

Service Set contains all replacement parts needed for one complete service.

One-headed pumps:

Service Set contains 1 diaphragm and 2 valve plates/sealings.

For pump type:	Order No.
N 820 AN.18	058081
N 820 AT.18	058078

Two-headed pumps:

Service Set contains 2 diaphragm, and 4 valve plates/sealings.

For pump type:	Order No.
N 820.3 AN.18	058080
N 820.3 AT.18	057358

10.2. Accessories

A program of accessories completes the LABOPORT® range of plumps:

Separator, High Performance Condenser, and Vacuum-Controller.

These components can quickly and easily be assembled to a system by simple plug-in connectors.

Base plate

Basic requirement for the assembly of a LABOPORT®-system is a base plate on which all components are mounted.

- Base plate and mounting:
Item **NP810/820**, Order No. **028129**

Separator

- Separator complete including tray:
Item: **NR 800** Order No. **026225**

High Performance Condenser

- High Performance Condenser complete including support
Item: **NH 800** Order No. **026231**

Vacuum-Controller

- Vacuum-Controller complete including support and Electrical Supply Unit
Item: **NC 800A** Order No. **046380**

- Vacuum-Controller complete without Electrical Supply Unit (for upgrading from system with one Vacuum-Controller to system with two Vacuum Controllers)

Item: **NC 800B** Order No. **046381**

- Two Vacuum-Controllers complete including support and Electrical Supply Unit

Item: **NC 800A/B** Order No. **046382**

Your KNF technical adviser will advise you how to upgrade your pump to a system. See page 1 for telephone number.

When ordering accessories please tell us to include the operating instructions for vacuum systems.

11. Returns

Pumps and systems used in laboratories and process-based industries are exposed to a wide variety of conditions. This means that the components contacting pumped media could become contaminated by toxic, radioactive, or otherwise hazardous substances.

For this reason, customers who send any pumps or systems back to KNF must submit a Health and safety clearance and decontamination form in order to avoid a hazardous situation for KNF employees. This Health and safety clearance and decontamination form provides the following information, among other things:

- physiological safety
- whether medium-contacting parts have been cleaned
- whether the equipment has been decontaminated
- media that have been pumped or used

To ensure worker safety, work may not be started on pumps or systems without a signed Health and safety clearance and decontamination form.

For optimal processing of a return, a copy of this declaration should be sent in advance via e-mail, regular mail, or fax to KNF Customer Service (refer to final page for address). In order to avoid endangering employees who open the shipment's packaging, despite any residual hazards, the original version of the Health and safety clearance and decontamination form must accompany the delivery receipt on the outside of the packing.

The template for the Health and safety clearance and decontamination form is included with these Operating Instructions and may also be downloaded from the KNF website.

The customer must specify the device type(s) and serial number(s) in the Health and safety clearance and decontamination form in order to provide for the unambiguous assignment of the Declaration to the device that is sent to KNF.

In addition to the customer's declaration of physiological safety, information about operating conditions and the customer's application are also of importance to ensure that the return shipment is handled appropriately. Therefore, the Health and safety clearance and decontamination form requests this information as well.

12. Health and safety clearance and decontamination form


 Form Rev. 02 / download: www.knf.com

Health and safety clearance and decontamination form

This declaration must be present and complete (the original must accompany the shipment's delivery receipt) before the returned device can be examined.

Device type:

Serial number(s):

.....

.....

Reason for returning the device (please describe in detail):

(The device(s) was(were) in operation ☐ yes ☐ no)

.....

.....

.....

.....

We confirm that the above device(s)

- ☐ has(have) pumped exclusively **physiologically unobjectionable** media and that it(they) are free of hazardous materials and any materials that are harmful to health.

Pumped media:

The device(s) was(were) cleaned ☐ yes ☐ no

- ☐ has(have) pumped media of the following category(categories) which are not physiologically unobjectionable and that cleaning of the device(s) (potentially only media-contacting parts) is required.

Name, chemical formula, Material Safety Data Sheet

☐ aggressive

☐ biological

☐ radioactive

☐ toxic

☐ other

The device(s) was(were) decontaminated and work can proceed without special measures ☐ yes

Method / proof:

.....

The device(s) was(were) not decontaminated and special measures are required before starting work ☐ yes

Measures:

.....

Legally binding declaration

We herewith affirm that the information provided in this form is correct and complete. Shipment of the devices and components is in compliance with statutory regulations.

.....
Company (stamp)

.....
Date

.....
Name

.....
Authorized signature

.....
Position

KNF worldwide

Please find our local KNF partners at: www.knf.com